Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/14/2005
PI Name:	Wong, Joyce Ph.D.		
Project Title:	Novel Microstructures for Vascular Tissue Engineering		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boston	**State:**	MA
Zip Code:	02215-2407	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-08-B
Start Date:	12/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (12/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Desai, Tejal (Boston University)		
Grant/Contract No.:	NNJ04HC73G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The major goal of this research project is to tissue-engineer small diameter blood vessel substitutes. Such tissue-engineered blood vessels can also serve as tissue models for studying the effects of microgravity environmental changes on tissue adaptation and to assess potential therapeutic strategies. Requirements for a blood vessel substitute include sufficient burst pressure, viscoelastic mechanical properties, vasoactivity, and nonthrombogenicity. Native blood vessels have evolved a sophisticated structure and hierarchical organization of cell types and matrix molecules in order to satisfy these requirements. Our central hypothesis is that the physiological spatial arrangement of cells and extracellular matrix in an engineered blood vessel will affect the viscoelastic mechanical properties, which in turn, will affect vascular smooth muscle and endothelial cell response. This hypothesis will be tested by answering questions directly posed in Specific Aims 1 and 2 in which the effects of two main biomaterial scaffold properties on tissue function will be explored: (1) microstructure and (2) mechanics. Our strategy of fabricating novel scaffolds with defined		

	microstructure and mechanical properties will help to reveal the effects of specific ultrastructural and mechanical factors on cell response and tissue development. We will use this approach to develop a new generation of biomaterials for vascular tissue engineering that will provide an in vitro model tissue in which parameters can be varied systematically to investigate cell mechanics, tissue development, cell-substrate and cell-cell interactions in controlled microgravity environments in order to attain a better understanding of effects of microgravity on human physiology. Furthermore, we recognize that these tissues may also be used as potential blood vessel replacements that are in high demand and are an integral part of more complex organs and tissues.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The ultimate goal of this research project is to create tissue-engineered small diameter vessels that can serve both as vessel replacements and as a model system that could be used to examine the physiological changes that occur during microgravity changes in space flight. Progress for this year has been in two areas. The first area has focused on examining the role of micropatterned and microtextured substrates on vascular smooth muscle cell functional response. These responses include contractility, protein expression and localization, and cell morphology. The second area has focused on developing a layered structure of the tissue-engineered vessel. Our main challenge has been to maintain adhesion between the various layers of the multi-layered constructs comprised of the biodegradable polymer PCL (poly caprolactone). We achieved this by developing polyethylene glycol (PEG)-dimethacrylate photopolymerizable polymers to serve as adhesive layers between the PCL layers. Using this system, we have been successful in maintaining cell viability in the multi-layered constructs. Ongoing studies are investigating whether the cellular response is altered in these multilayered constructs.
Bibliography Type:	Description: (Last Updated: 01/09/2014)
Articles in Peer-reviewed Journals	Jacot JG, Dianis S, Schnall J, Wong JY. "A simple microindentation technique for mapping the microscale compliance of soft hydrated materials and tissues." Journal of Biomedical Materials Research, in review, October 2005. , Oct-2005
Articles in Peer-reviewed Journals	Sarkar S, Lee GY, Wong JY, Desai TA. "Development and characterization of a porous micro-patterned scaffold for vascular tissue engineering applications." in preparation, Nov 2005. , Nov-2005
Articles in Peer-reviewed Journals	Köster S, Leach JB, Struth B, Pfohl T, Wong JY. "Visualization of flow-aligned type I collagen self-assembly in tunable pH-gradients." PNAS, in review, Nov 2005. , Nov-2005
Articles in Peer-reviewed Journals	Sarkar S, Dadhania M, Rourke P, Desai TA, Wong JY. "Vascular tissue engineering: microtextured scaffold templates to control organization of vascular smooth muscle cells and extracellular matrix." Acta Biomaterialia 2005 Jan;1(1):93-100. PMID: 16701783 , Jan-2005
Dissertations and Theses	Jacot JG. "Multiscale analysis of arterial bypass remodeling: Mechanical characterization and influence of endothelial injury on vascular smooth muscle cell proliferation." PhD Thesis, Boston University, 2005. , May-2005
Dissertations and Theses	Foster MD. "Microgrooves dominate cyclic strain response of vascular smooth muscle cell orientation." MS Thesis, Boston University, July 2005. , Jul-2005
Patents	U.S. provisional patent. U.S. provisional patent, November 2005. Nov-2005 Wong JY, Desai TA. "Novel Microstructures for Vascular Tissue Engineering."
Presentation	Williams, C.; Brown, X.Q.; Wong, J.Y. "Controlling Vascular Smooth Muscle Cell Phenotype Through Substrate Topography" North East Smooth Muscle Society 12th annual meeting, Worcester, MA Oct-2004
Presentation	Williams, C.; Brown, X.Q.; Wong, J.Y. "Controlling Vascular Smooth Muscle Cell Phenotype Through Substrate Topography" BEACON Bionanotechnology Symposium and Fair, Hartford, CT Oct-2004
Presentation	Jacot, J.; Jackel, J.L.; Koester, S.F.; Wong, J.Y. "A photolithographic method for patterning soft polyacrylamide to enhance smooth muscle cell elongation" AVS Science & Technology Fall Meeting, Boston MA Nov-2005
Presentation	Jacot, J.; Wong, J.Y. "An endothelial injury front stimulates smooth muscle cell proliferation in highly localized regions" Biomedical Engineering Society Fall Meeting, Baltimore, MD Sep-2005
Presentation	Sarkar, S.; Leach, J.B.; Wong, J.Y.; Desai, T.A. "Fabrication of a Highly Organized 3-D Tissue Construct for Vascular Tissue Engineering" Biomedical Engineering Society, Fall Meeting, Baltimore, MD Sep-2005
Presentation	Williams, C.; Brown, X.Q.; Ma, H.; Chilkoti, A.; Wong, J.Y. "Critical Constraints: Regaining Vascular Smooth Muscle Contractility via Microtopographical Cues" Biomedical Engineering Society Fall Meeting, Baltimore MD Sep-2005
Presentation	Leach, J.B.; Koester, S.; Pfohl, T.; Wong, J.Y. "Microaligned collagen matrices: Biomimetic microstructured materials from natural biopolymers" Biomedical Engineering Society, Fall Meeting, Baltimore MD Sep-2005
Presentation	Leach, J.B.; Koester, S.; Pfohl, T.; Wong, J.Y. "Microaligned collagen matrices: Biomimetic microstructured materials from natural biopolymers" AIChE (American Institute of Chemical Engineers) Fall Meeting, Cincinnati, OH Nov-2005